

ESP-OS Installation Guide

This guide explains how to install and run ESP-OS on ESP32.

1) Requirements

Hardware

- ESP32 dev board
- MicroSD card (recommended FAT32)
- USB data cable
- Optional: 3.5" ESP32 display module

Software

- [PlatformIO](https://platformio.org/) (<https://platformio.org/>) via:
- VS Code extension, or
- CLI (`pip install -U platformio`)
- USB serial driver for your ESP32 board (if needed)

2) Get the source code

```
git clone https://github.com/0kurka1/esp-os-lua.git
cd esp-os-lua
```

3) Prepare SD card

1. Format SD card as FAT32.
2. Copy folder `sd_card_files/ESP-OS/` to SD card root.
3. Verify key files exist on SD card:
 - `/ESP-OS/boot/init.lua`
 - `/ESP-OS/system/kernel.lua`
 - `/ESP-OS/config/system.lua`
 - `/ESP-OS/config/wifi.lua`

4) Build firmware

```
pio run
```

5) Upload firmware

```
pio run -t upload
```

If your board is not detected, run:

```
pio device list
```

Then configure upload port in `platformio.ini` if needed.

6) Open Serial Monitor

```
pio device monitor -b 115200
```

Expected boot flow:

1. POST checks are printed.
2. SD mount check passes.
3. Lua kernel starts.
4. Main menu appears.

7) First Run Tips

- Use single-key input in menu (`1..5` , `Q` , `B`) where applicable.
- Wi-Fi passwords use masked input (`serial.readPassword`).
- If SD is unavailable, some features run in limited mode.

8) Common Problems

SD card not detected

- Reformat FAT32.
- Check SD wiring and CS pin.
- Verify `SD_CS_PIN` in `include/config.h`.

Upload fails

- Hold BOOT button while flashing (some boards).
- Check cable (must support data, not only power).
- Close other serial tools using the same COM port.

Garbage serial output

- Ensure monitor baud is `115200`.

Lua script error at boot

- Verify files on SD card and paths.
- Check syntax in modified Lua scripts.

9) Version Reference

Current target release: **v0.0.3.1**

See [CHANGELOG.md](#) (CHANGELOG.md) for full release history.